

IT 630: Data Science and Scalable Data Systems

Fall 2026 · 2026-08-31 to 2026-12-14 · 4 credits

Course Catalog

A practical course on building the data foundation for data science and analytics. Create reliable, scalable data for dashboards, apps, and machine learning—with an introduction to core concepts in data modeling, pipeline design, quality assurance, streaming, and governance. Prepares students to deliver dependable data that powers real analysis and products—skills in high demand across careers in analytics, AI, and machine learning.

Credits: 4

Prerequisites

ADMN 520, BIOL 528, BUS 430, HMP 540, MATH 439, MATH 539, MATH 540, MATH 644, or PSYC 402.

Course Designation

undergraduate

University Requirements

Required for Analytics and Data Science majors. Elective for Computer Science and Information Technology majors, and for Data Science and Analytics minors. Not writing intensive.

Student Outcomes

By the end of this course, students should be able to:

- Apply theory, techniques, and tools throughout the data science lifecycle and employ the resulting knowledge to satisfy stakeholders' needs.
- Acquire data from files, APIs, and web sources.
- Transform messy, nested, or incomplete data into usable analytical structures.
- Use SQL and modern data tools to query, validate, and explain data.
- Diagnose why data analyses disagree, including grain, missingness, leakage, denominator errors, and definition drift.
- Build data artifacts that are reproducible, documented, and testable.
- Explain how curated data supports dashboards, machine learning datasets, and AI-enabled applications.
- Use AI tools as collaborators while verifying and taking responsibility for the final work.

Letter-grade Scale

UNH Letter Grade By Percentage

Grade	Range	Grade	Range
A	100% to 94%	A-	< 94% to 90%
B+	< 90% to 87%	B	< 87% to 84%
B-	< 84% to 80%	C+	< 80% to 77%
C	< 77% to 74%	C-	< 74% to 70%
D+	< 70% to 67%	D	< 67% to 64%
D-	< 64% to 60%	F	< 60% to 0%

Credit Hour Workload Estimate

This is a 4-credit, 15-week course. Per UNH policy, students should expect approximately 180 hours of academic work across the term (45 hours per credit), or about 12 hours per week. This is a 4-credit, 15-week course. The 180-hour total includes approximately 35 contact hours across the Monday and Wednesday lectures and the Friday applied activity, with the remainder as structured outside work. Weekly work includes lecture preparation, readings or walkthroughs, the Friday applied activity, assignments, short quizzes, and incremental progress on the group project.

Topics

Course frame and the data lifecycle

- The data lifecycle, and where two defensible analyses diverge
- Reproducibility, project repositories, and traceable results
- Data ethics and what a claim can responsibly assert

Data acquisition

- Files and formats, and what each format hides
- APIs, the request/response model, authentication, rate limits, pagination
- Nested data to tabular structure; scraping as a last resort
- Acquisition provenance as the first line of a data card

Data modeling for analytics

- Entity and grain - what exactly one row represents
- Keys as promises the data may not keep
- Fact and dimension tables, star and snowflake schemas, wide versus narrow

Structured data and SQL foundations

- Filtering, projection, and grouping that changes the grain
- Joins, one-to-many row inflation, nulls, and three-valued logic
- Window functions, and why every rate needs a stated denominator

Files, databases, warehouses, and lakehouse concepts

- Row versus columnar storage; Parquet and query performance
- Transaction logs, ACID table concepts, and schema evolution
- Warehouse, lake, and lakehouse as design choices

Transformations and pipeline patterns

- ETL versus ELT; staging/bronze, clean/silver, curated/gold contracts
- Idempotence and safe re-runs
- The pipeline step as a reviewable artifact

Data quality, testing, and validation

- Constraints and expectations as checks that run and fail loudly
- Anomaly checks against history; reject, quarantine, or repair as policy
- The data card - what the data can and cannot support

Incremental data and slowly changing dimensions

- Full versus incremental refresh; detecting what changed at the source
- Merge and upsert
- SCD1 versus SCD2, and which questions each makes unanswerable

Streaming fundamentals

- Batch and stream as a continuum; event time versus processing time
- Windows over unbounded data and watermarks as a bet about lateness

Performance and cost basics

- Partitioning, statistics, and how an engine avoids reading data
- Skew, broadcast versus shuffle, caching correctness, and the cost tradeoff

Analytics at scale

- Advanced window patterns and time-series joins
- Rollups, aggregate hierarchies, and BI-ready output with a documented grain

Data for machine learning

- Feature tables versus label tables
- Leakage, time-based splits, and point-in-time correctness
- The model families those features feed

Serving and access patterns

- Batch extracts, query endpoints, and the semantic layer
- Service-level documentation for a data product

Governance, privacy, and reproducibility

- Lineage, run logs, and PII handling
- Documentation and accountability over everything already built

Agents and AI-enabled data workflows

- LLMs with tools in a loop; a tool call as request/response with the caller swapped
- Structured outputs as a contract; retrieval over curated data
- Evaluations and guardrails when the author of a query is not human

Applications and integration

- Dashboards and supervised ML as two end-to-end consumers of the same curated data
- Problem framing revisited, and capstone integration

Sections

Modality: Hybrid

Section 02, CRN 13098. MWF 11:10 a.m.–12:00 p.m., Kingsbury N129, Durham.

Course Format

Monday and Wednesday meetings are typically in person. Friday is normally a virtual learning activity. The specific Friday activity can vary by week.

Faculty

Matt Magnusson, Senior Lecturer, Computer Science

Preferred contact: email. Email: mfm2@unh.edu. Response time: within 24 hours on weekdays.

Office: Kingsbury W243. Phone: 603-862-3794.

Office hours: Monday and Wednesday 9:00–11:00 a.m., or by appointment (Kingsbury W243 or Teams).

Evaluation

Component	Percentage
Quizzes	10%
Midterm examination	15%
Final examination	15%
Group projects	25%
Assignments	20%
Prepared participation and practice	15%

How You Will Be Assessed

Quizzes

Types: Multiple choice, multiple answer, short answer, fill-in-the-blank, and interpretation of small data, code, or query examples.

Delivery: In person during a scheduled class period, or MyCourses when announced.

Quizzes reinforce recent material and offer quick feedback before larger assessments. They focus on concepts, vocabulary, and diagnostic reasoning rather than on production of large artifacts. Quizzes are taken at the scheduled time. Make-ups are not normally provided; if a documented conflict or illness arises, email the instructor before the deadline when possible.

Midterm examination

Types: Written examination; may include multiple choice, short answer, and applied problems.

Delivery: In person during a scheduled class period.

One midterm examination during the semester. The exam assesses your individual understanding of concepts and skills covered up to that point. Format and coverage will be announced in class. University policy prohibits scheduling exams during the last week of classes.

This offering: Monday, October 19. Covers the course frame, data acquisition, data modeling, SQL, and the files-to-tables unit, plus the Friday applied activity: acquire, shape, query, store.

Final examination

Types: Written examination; may include multiple choice, short answer, and applied problems.

Delivery: In person during the scheduled final-examination period.

Cumulative final examination during the university final-examination period. The exam assesses individual understanding across the course. Format and coverage will be announced in class.

The final assesses your ability to connect course concepts across the data lifecycle: acquisition, modeling, querying, quality, pipelines, analytics, machine learning data, and AI-enabled data workflows.

Group projects

Types: Collaborative data artifacts with documentation, plus individual contribution evidence through participation, documentation, and peer feedback.

Delivery: MyCourses, GitHub, or another specified platform, with in-class presentation or cross-review when scheduled.

Group projects provide hands-on experience working collaboratively to solve real-world problems. They emphasize teamwork, communication, and integration of diverse skills to design, build, and evaluate a trustworthy data artifact. Grades reflect both collective outcomes, such as the quality, rigor, and creativity of the final product, and individual contributions demonstrated through participation, documentation, and peer feedback. The project is built incrementally across the term rather than started in the final weeks.

Assignments

Types: Applied labs, individual data-work submissions, and reflection or verification notes where assigned.

Delivery: MyCourses unless specified otherwise, following any instructions on file format and naming.

Assignments are hands-on applications of course concepts. You may work with files, APIs, SQL, notebooks, data validation checks, data cards, dashboards, ML-ready datasets, or small AI and data workflows. Work is due at the stated deadline. All submissions must follow course and university academic integrity policies; cite sources and collaborators as required.

Prepared participation and practice

Types: Assigned readings, discussion boards, classroom activities, polls, and brief reflections.

Delivery: In class or through MyCourses, as assigned.

Class participation includes assigned readings, discussion boards, classroom activities, polls, and brief reflections. Credit is based on completing a genuine attempt; simply attending is not enough.

This offering: in-class activities during Monday and Wednesday lectures, the Friday applied activity, quick checks, discussions, short prework, and occasional self-learning tasks. The Friday activity is scheduled course work and counts in this category. Participation activities cannot be made up for credit.

Capstone Project

The group project is the culminating artifact of the course. Each team builds a trustworthy data artifact from a real or realistic data source, using a course-provided dataset, an approved team-selected dataset, or another approved source. Every project should include curated data tables or equivalent structured artifacts, basic quality checks or validation evidence, a reproducible query, report, or workflow, a data card explaining what claims the data can and cannot support, and a final explanation of what can be trusted, what cannot, and why. A team may emphasize one depth area: analytics and BI, machine learning dataset preparation, pipeline design, inference and claims, or an AI-enabled data workflow. The project is built incrementally during the course; the final weeks are for integration, defense, and cross-review, not for starting from scratch.

Reflection / Verification Notes

Because AI tools are allowed and because the course assesses judgment, students regularly submit short verification or reflection notes. These notes should answer: What did I check? What could still be wrong? Why do I trust this result enough to submit it? Where did AI help, if it did? Verification is part of the course outcome, not optional decoration.

Late Policy for Assignments

Assignments must be completed by the posted due date unless otherwise specified. If a non-academic circumstance such as extended illness, family leave, or another serious situation affects your ability to complete work, contact the Dean of Students. They can verify your situation to instructors while protecting your privacy and helping flexibility be applied fairly.

Temporary academic supports

If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, you may reach out to the Dean of Students (dean.students@unh.edu) to request a letter be sent to all your faculty.

If you are required to miss significant class time, you will be provided temporary academic supports so that you can continue to make satisfactory progress in this course. Please contact your course instructor to discuss the specific types of supports that will be implemented during your absence. Possible options you may be provided in this class include:

- Class notes from a peer
- One-on-one meetings with instructor or TA to catch-up on missed content
- Virtual office hours
- Access to recorded lectures (asynchronous)
- Remote access for class (synchronous)

- Lecture slides
- Handouts or other materials that are distributed

Please recognize that not all students will be well enough to attend classes remotely.

Grade Disputes

If you believe a grade on an assignment, quiz, or exam is incorrect, submit a written regrade request within one week of the grade being posted to MyCourses. Coursework submitted for reconsideration may be regraded in its entirety.

Learning Resources

Textbook

No required textbook. Readings required for the course will be provided through MyCourses.

Other materials

None

Technical requirements

A laptop is required for in-class work and for the Friday applied activity. Using a smartphone for these activities is not sufficient. All course tooling runs in the browser; no local installation and no paid account are required. If you do not have a laptop, contact the instructor so arrangements can be made.

Required Software: Google Colab

This course uses Google Colab for some activities and assignments. Colab is required but free: you need a Google account to access it. You do not need a paid Colab Pro or Pro+ subscription for this course.

Attendance

Attendance is expected at all scheduled meetings, including the Friday applied activity. In-class participation activities cannot be made up for credit if missed. Prepared contribution is required; presence alone is insufficient.

Student AI Policy

AI tools are allowed in this course and will be part of how we learn applied data work. Students may use AI tools to help explain concepts, draft code, write SQL, debug errors, interpret documentation, brainstorm approaches, and improve written explanations. However, students are responsible for everything they submit. Allowed use includes asking AI to explain an error or concept; asking AI to draft code or SQL that you then test and revise; asking AI to help summarize documentation; asking AI to suggest checks, tests, or edge cases; and asking AI to improve clarity of writing. Not allowed: submitting AI-generated work you cannot explain; submitting unchecked AI output as fact; fabricating data, citations, sources, or results; using AI to bypass required individual work on quizzes or assessments; hiding meaningful AI assistance when disclosure is requested. The working standard is: AI may help you produce the work, but you must verify it, understand it, and take responsibility for it.

Instructor AI Policy

I use AI tools to help develop this course, including lectures, assignments, exercises, examples, and supporting materials. AI helps me iterate on explanations, generate examples, test assignment ideas, and improve the clarity of course materials. I grade student assignments personally. I may use AI as a tool to help improve the quality and consistency of feedback, such as identifying patterns, suggesting clearer explanations, or helping ensure feedback is thorough. Final evaluation and grades are my responsibility and my judgment. This is the same standard expected of students: AI can assist the work, but the person using it remains responsible for the result.

Academic Honesty

The University Academic Integrity Policy applies in this course. Students are expected to complete independent, original work unless the instructor specifies otherwise. The policy, including the steps faculty must take when issuing a formal penalty (reduced or failing grade on an academic assessment or examination, or reduced or failing grade in the course), is published in the Student Rights, Rules, and Responsibilities. Those steps include providing students with notice of the allegations, giving them the opportunity to respond, and reporting the violation to the Office of Community Standards if a formal grade penalty is being issued.

See the [Academic Integrity Policy](#) and [Academic Integrity in the Student Rights, Rules, and Responsibilities](#).

Plagiarism: Use or submission of intellectual property, ideas, evidence produced by another person, including computer generated text or work outsourced to third parties, in whole or in part as one's own in any academic assessment without providing proper citation or attribution. In some cases, reusing one's own previous work without acknowledging or citing the original work can constitute self-plagiarism.

Cheating: Use or attempted use of any academic exercise materials, information, study aids, electronic data, AI tools, assignment/exam surrogate, or other forms of assistance without authorization.

Class Recording

Class lectures and activities may be recorded to make this course more accessible, and recordings may be made available to students presently enrolled in the class to assist in reviewing material and preparing for assessments. Recordings are to be used for educational purposes only. You may not record class activities (including audio/photographs/screenshots of instructors and students) without express permission from me and other students whose image is captured (See NH RSA-570A). If permission is granted, student-initiated recordings may only be used for educational purposes in this class and only during the period in which you are enrolled in the class; authorized student-initiated recordings may not be posted or shared in any fashion outside of the class.

University Policies

UNH provides a variety of resources to support students' academic success, health and well-being, and engagement with the campus community.

A list of student resources can be found on the [Syllabus Supplemental Resources](#) page. These resources are not required for this course but may be helpful to you during the semester.

Students with Disabilities

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) at sas.office@unh.edu.

Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS, but are under no obligation to provide retroactive accommodations.

For more information, contact SAS: www.unh.edu/sas, (603) 862-9355, 711 (Relay NH) or sas.office@unh.edu.

Students Experiencing Emotional or Mental Health Distress

Your academic success and overall mental health are very important. If, during the semester, you find you are experiencing emotional or mental health issues, please contact Psychological and Counseling Services (PACS), which provides counseling appointments and other mental health services.

Location: The Wellness Center, 4 Pettee Brook Lane. Phone (24/7): 603.862.2090 / TTY: 7-1-1.

If urgent, students may call PACS Monday–Friday, 8 a.m.–5 p.m. and schedule an urgent same-day appointment.

Students Experiencing Housing or Food Insecurity

Food, housing, and financial supports are available on the Durham campus. See Housing, Food and Financial (Basic Needs) Support. If you have questions or need to speak with someone, contact basic.needs@unh.edu or visit the Memorial Union Building.

Classroom Behavior Expectations

Understanding your rights and responsibilities is essential to fostering a respectful and inclusive campus climate, including in the classroom. The Student Rights, Rules, and Responsibilities Handbook (“S.R.R.R.”) is the University’s official compendium for the Student Code of Conduct, cornerstone student policies and expected standards of behavior. See Academic Integrity in the S.R.R.R. and the Syllabus Supplemental Resources institutional-policies list.

Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university’s Title IX Coordinator (Bo Zarycky, Bo.Zarycky@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy at (603) 862-7233 / TTY (800) 735-2964.

For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, and your rights and reporting options at UNH (including anonymous reporting options), see the Civil Rights & Equity Office. The uSafeUS® app is your college’s free, anonymous tool that helps you and your friends navigate uncomfortable situations and access support.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or (603) 862-2930 voice / (603) 862-1527 TTY / 7-1-1 Relay NH. Anonymous reports may be submitted.